SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1507

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 20**
Question Count: 5

Code of Conduct

I Lakshana K [192524433] (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: ____________________

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

Q1 (10 Marks): A retail company wants different business applications to communicate seamlessly. Analyse how Service-Oriented Architecture (SOA) can solve this problem.

Answer:

Service-Oriented Architecture (SOA) is a software architecture in which different business functions are designed as independent and reusable services. These services communicate with each other using standard protocols, allowing different applications to work together efficiently. SOA helps organizations integrate applications developed using different programming languages, operating systems, and databases.

In a retail company, several applications such as Inventory Management, Billing System, Customer Relationship Management (CRM), Order Processing, Payment Gateway, Warehouse Management, and Shipping System are used daily. Without SOA, these applications operate independently, making data sharing difficult and increasing processing time. SOA solves this problem by enabling these applications to exchange information seamlessly.

For example, when a customer places an online order, the Order Processing Service sends a request to the Inventory Service to check product availability. If the item is available, the Payment Service processes the payment. After successful payment, the Warehouse Service prepares the product for dispatch, and the Shipping Service updates the delivery details. At the same time, the CRM Service updates the customer's purchase history. All these services communicate automatically without manual intervention.

Advantages of SOA in a Retail Company:

- Enables seamless communication between different business applications.
- Supports interoperability between different technologies and platforms.
- Services are reusable, reducing development time and cost.
- Easy to maintain because each service works independently.
- Improves scalability by allowing new services to be added without affecting existing ones.
- Enhances customer satisfaction through faster order processing.
- Reduces maintenance and integration costs.

Conclusion:

Service-Oriented Architecture is an effective solution for integrating different retail applications. It improves communication, increases flexibility, reduces operational costs, and ensures faster and more reliable business operations, making it an ideal architecture for modern retail organizations.

Q2 (10 Marks): Explain how web services enable SOA in cloud computing.

Answer:

Web services are software components that allow different applications to communicate and exchange data over the Internet using standard protocols such as HTTP. They are the foundation of Service-Oriented Architecture (SOA) because they enable different services to interact regardless of the programming language or platform used.

In cloud computing, applications are hosted on remote cloud servers. Web services allow these applications to communicate with one another through standardized interfaces called APIs. The two commonly used web service technologies are SOAP (Simple Object Access Protocol) and REST (Representational State Transfer). SOAP mainly uses XML for data exchange, whereas REST commonly uses JSON, making it lightweight and faster.

For example, an online shopping application hosted in the cloud uses multiple web services. The Authentication Service verifies the user login, the Payment Service processes online payments, the Inventory Service checks product availability, and the Shipping Service provides delivery updates. These independent services work together through web services, ensuring smooth communication and efficient business operations.

Role of Web Services in SOA:

- Provides standard communication between different applications.
- Supports platform and programming language independence.
- Enables easy integration of cloud-based and third-party applications.
- Promotes service reusability across multiple systems.
- Improves scalability and flexibility in cloud environments.
- Simplifies application maintenance and updates.
- Provides secure and reliable data exchange.

Conclusion:

Web services are the key technology that enables SOA in cloud computing. They provide a standardized way for applications to communicate, making cloud systems more flexible, scalable, interoperable, and efficient. As a result, organizations can easily integrate different services and deliver better cloud-based solutions.